COE304-F01

College of Engineering & Computer Science

Computer Engineering Department

The smart Emergency vehicle

Embedded Systems [COE-304]

جامعة المستقبل
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ABSTRACT

This project presents a wireless gold detection vehicle designed for safe exploration in hazardous areas. The system integrates a hand-wound coil for metal detection, Arduino-based control, and Bluetooth communication for remote operation. Built on a 4WD chassis, it alerts users via buzzer and LED when gold is detected, offering a safer and more efficient solution for mineral exploration.

INTRODUCTION

Traditional gold mining methods expose workers to serious health risks, with Colombia emitting over 150 tons of mercury annually and 50% of gold workers in Burkina Faso reporting toxic symptoms. Additionally, mining causes severe environmental damage, with 1.32 kg of mercury lost for every 1 kg of gold extracted in regions like Tanzania. As global demand for metal detection technology is projected to double by 2033, this project introduces a wireless, Arduino-based gold detection vehicle. It enables safer, faster exploration by remotely locating gold in hazardous areas without risking human lives.

OBJECTIVES

1. Design and implement an efficient and cost-effective system for gold exploration, reducing operational expenses while enhancing detection capabilities.

2. Develop and offer a remote-controlled vehicle that ensures safe and secure access to hard-to-reach areas, minimizing human risk and expanding exploration possibilities.

3. Establish and integrate autonomous navigation capabilities for the vehicle to automatically detect gold-rich areas, improving efficiency and reducing the need for manual intervention.

MATERIALS & METHODS

- **Arduino:** The Arduino Uno and Nano microcontrollers were used to control the vehicle's movement, process signals from the gold detection circuit, and manage wireless Bluetooth communication with the operator.
- **Copper Coil:** A hand-wound copper coil was constructed to generate electromagnetic fields, allowing the detection of gold based on changes in conductivity when a metal object is present.
- **Vehicle Body:** The entire system was mounted on a 4WD LAFVIN car chassis, providing stability, flexibility, and mobility over rough mining terrains while supporting all motors, sensors, and control boards.

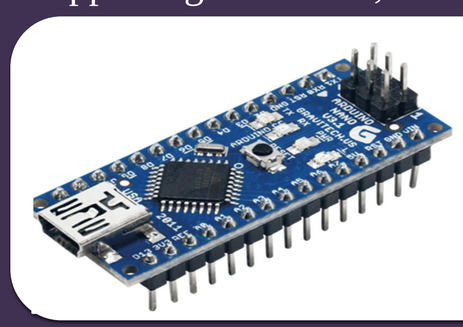

Figure 1. Arduino microcontrollers.

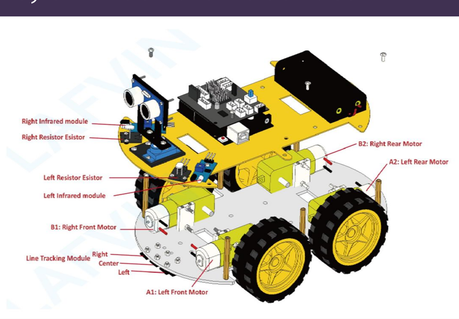

Figure 2. Vehicle Body.

Figure 3. Copper Coil.

RESULTS & DISCUSSION

The gold detection vehicle successfully achieved wireless control via the LightBlue Explorer app, allowing smooth navigation and real-time metal detection. The system effectively identified gold-like objects using electromagnetic sensing, providing immediate alerts through buzzer and LED signals. Several challenges were encountered, including PCB printing issues, component burnout, electromagnetic interference, and messy wiring; however, they were resolved through circuit improvements, better shielding, and careful system organization. Overall, the project met its goals by delivering a safe, remote-controlled exploration tool capable of detecting metals in hazardous environments with high reliability.

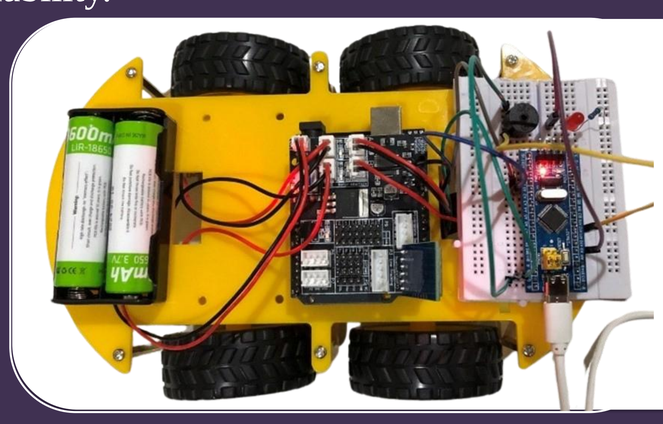

Figure 1. Final Project.

Figure 2. Vehicle Body Implementation.

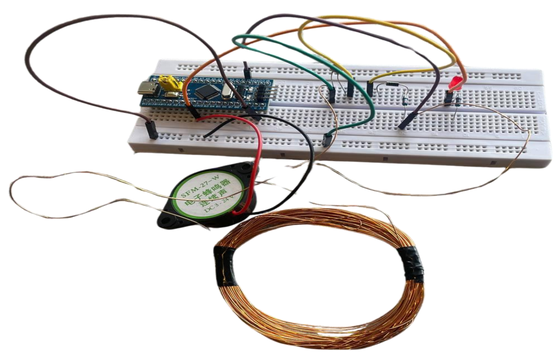

Figure 3. Gold Detector Circuit Implementation.

CONCLUSION

A wireless gold detection vehicle was developed using embedded systems to explore hazardous areas safely. With Arduino control, Bluetooth communication, and a custom detection circuit, the system achieved reliable performance. Challenges like interference and wiring issues were solved, resulting in a safer and more efficient tool for modern mineral exploration.

REFERENCES

[1] S. Al Mutairi, S. Al Sultan, A. AlAnzy .2025. "Gold Detection Vehicle ". Mustaqbal University, Saudi Arabia.